

Cycle Graphs Exceed the Conjectured Cycle-Count Bound: A Disproof of Conjecture 00000004413

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Abstract

The source asserts that in a graph limit with degree bound d the measure of cycles of length ℓ is at most $(d-1)^\ell/(2\ell)$, and that this bound is attained and optimal for the d -regular tree. For $d=2$ the bound is false: the cycle graph C_ℓ has degree 2 and cycle measure $1/\ell > 1/(2\ell)$, for every $\ell \geq 3$. This is verified in Lean for Mathlib's graphs. For $d \geq 3$ the bound holds but is neither attained nor optimal, and for no $d \geq 2$ does the tree attain it; these two facts are proved here on paper. An earlier submission with the triangle as example was withdrawn by its author; it is discussed below.

Reading of the source

A bounded-degree graph limit (Benjamini–Schramm limit, or graphing) is a random rooted graph (G, o) with all degrees at most d that satisfies the mass-transport principle. Every finite graph G with a uniformly random root is such an object; it is the limit of the constant sequence $G, G, \dots$. For a finite graph the *measure of cycles of length ℓ* is

$$c_\ell(G) = \frac{\#\{\text{cycles of length } \ell \text{ in } G\}}{|V(G)|},$$

and for a general limit it is $\mathbb{E}[N_\ell(G, o)]/(2\ell)$, where $N_\ell(G, o)$ is the number of sequences $(o = v_0, v_1, \dots, v_{\ell-1})$ of distinct vertices with v_i adjacent to v_{i+1} for all i modulo ℓ . The two agree for finite graphs, because each cycle of length $\ell \geq 3$ gives exactly 2ℓ such sequences in total, one for each starting vertex and direction. A cycle is a cycle in the graph-theoretic sense: its vertices are distinct.

The source makes two claims for a degree bound d and a length ℓ :

- (i) $c_\ell \leq (d-1)^\ell/(2\ell)$ for every graph limit with degree bound d ;
- (ii) this bound is attained, and is optimal, for the d -regular tree.

Other normalisations of the measure are larger: the expected number of ℓ -cycles through the root is ℓc_ℓ . So a violation of (i) for c_ℓ is a violation for them as well.

The disproof

Theorem 1. *Let $\ell \geq 3$. The cycle graph C_ℓ has all degrees equal to 2 and $c_\ell(C_\ell) = 1/\ell$. Hence*

$$c_\ell(C_\ell) = \frac{1}{\ell} > \frac{1}{2\ell} = \frac{(2-1)^\ell}{2\ell},$$

and claim (i) is false for $d=2$ and every $\ell \geq 3$.

Proof. Take the vertex set $\mathbb{Z}/\ell$ with i adjacent to $i \pm 1$. Since $\ell \geq 3$, the two neighbours $i - 1$ and $i + 1$ of i are different, so every degree is 2. The graph C_ℓ is itself a cycle of length ℓ , and it is the only one: a cycle of length ℓ has ℓ edges and C_ℓ has only ℓ edges. So $c_\ell(C_\ell) = 1/\ell$.

For the formal proof only the inequality $c_\ell(C_\ell) \geq 1/\ell$ is needed. The ℓ rotations $i \mapsto i + k$ and the ℓ reflections $i \mapsto k - i$ are injective homomorphisms $C_\ell \rightarrow C_\ell$. They are 2ℓ different maps: a rotation is determined by the image k of 0, a reflection likewise, and if $i + k = k' - i$ for all i , then $k = k'$ and $2 = 0$ in $\mathbb{Z}/\ell$, which is false for $\ell \geq 3$. So there are at least 2ℓ rooted oriented cycles and $c_\ell(C_\ell) \geq 2\ell/(2\ell \cdot \ell) = 1/\ell$. $\square$

Proposition 2. *Let (G, o) be a rooted graph with all degrees at most d , where $d \geq 1$, and let $\ell \geq 3$. Then $N_\ell(G, o) \leq d(d-1)^{\ell-2}$. Hence every graph limit with degree bound d satisfies*

$$c_\ell \leq \frac{d(d-1)^{\ell-2}}{2\ell}.$$

For $d \geq 3$ this is at most $\frac{3}{4} \cdot \frac{(d-1)^\ell}{2\ell}$.

Proof. Build a sequence $(o = v_0, \dots, v_{\ell-1})$ step by step. There are at most d choices for v_1 . For $2 \leq i \leq \ell - 1$ the vertex v_i is a neighbour of v_{i-1} different from v_{i-2} , so there are at most $d - 1$ choices. This gives at most $d(d-1)^{\ell-2}$ sequences. Taking expectations and dividing by 2ℓ gives the bound for c_ℓ . Finally $d(d-1)^{\ell-2} = \frac{d}{(d-1)^2}(d-1)^\ell$, and $d/(d-1)^2 \leq 3/4$ for $d \geq 3$. $\square$

Claim (ii) fails for every $d \geq 2$. A tree has no cycle, so the d -regular tree has $c_\ell = 0$, while $(d-1)^\ell/(2\ell) > 0$ for $d \geq 2$. So the tree does not attain the bound. The bound is not optimal either: for $d = 2$ it is false by Theorem 1, and for $d \geq 3$ no graph limit comes within the factor $3/4$ of it, by Proposition 2. For $d = 2$ the correct optimal bound is $1/\ell$, attained by C_ℓ , again by Proposition 2.

Remark. The expression $(d-1)^\ell/(2\ell)$ is the limiting mean of the total number of ℓ -cycles in a random d -regular graph on n vertices as $n \rightarrow \infty$. It is a count for the whole graph, not a density, and not an upper bound. The refutation of (i) uses $d = 2$ only; for $d \geq 3$ claim (i) is true by Proposition 2, so the conjecture fails for $d \geq 3$ through claim (ii) alone.

The earlier submission

Pull request 302 of the competition repository (by `orionsheep`) treated this conjecture with the triangle K_3 , which is the case $\ell = 3$ of Theorem 1. It was closed without merging on 3 October 2026, together with a comment of its author withdrawing the submission; its comment thread contains no reviewer verdict. Its example and its conclusion were correct.

We read its Lean file. It modelled a finite graph by a structure of its own and defined the cycle measure through maps from $\mathbb{Z}/n$ to the vertices with cyclically adjacent consecutive values, identified by their edge sets. These maps were not required to be injective. For $n = 3$ they are exactly the triangles, so the instance used was sound. For $n \geq 4$ they include closed walks that are not cycles, such as a, b, a, b on a single edge; with that definition a single edge would have positive “4-cycle measure”. So the universally quantified statement negated there was not the bound for cycles in general. Its statement about the tree was only the positivity of the expression $(d-1)^\ell/(2\ell)$.

The present submission uses Mathlib’s graphs and cycle graphs, counts genuine cycles (injective homomorphisms from C_ℓ), proves the violation for every $\ell \geq 3$, and separates what holds for $d \geq 3$.

Formal verification and scope

Graphs are Mathlib’s `SimpleGraph`; degrees are Mathlib’s `SimpleGraph.degree`; the cycle graph is Mathlib’s `SimpleGraph.cycleGraph`. The type `CycleCopy G L` consists of the injective graph homomorphisms from `cycleGraph L` to `G`, and

`cycleMeasure G L`

is their number divided by $2L$ and by the number of vertices, a rational number. `claimedBound d L` is $(d - 1)^L / (2L)$. `rotation` and `reflection` are the maps of the proof of Theorem 1, and `two_mul_le_card` shows that they are $2L$ different elements. The theorem

`cycle_violates`

states that `claimedBound 2 (1 + 3)` is smaller than the cycle measure of `cycleGraph (1 + 3)` for every natural number l .

The proposition `ClaimedUpperBound` is claim (i) for all finite graphs, all d and all $L \geq 3$. Its negation is

`conjecture_false,`

and `conjecture_false_every_length` gives, for each $L \geq 3$, a graph with all degrees at most 2 violating the bound for $d = 2$. No `sorry`, `native_decide` or custom axiom occurs.

Scope. Lean treats finite graphs, which are special graph limits; general graph limits are not formalised, and a counterexample among finite graphs refutes claim (i) for graph limits. Lean proves $c_\ell(C_\ell) \geq 1/\ell$; the equality, Proposition 2, the discussion of claim (ii), and the remark are proved on paper only. The refutation in Lean does not use the degenerate value $d = 0$, for which the expression $(d - 1)^L / (2L)$ is negative when L is odd.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000004413.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.